VISVESVARAYA TECHNOLOGICAL UNIVERSITY, BELAGAVI

A project report
on

"IoT Based Safety Helmet for Coal Minors"

submitted in partial fulfillment of the requirements for the award of the degree of

BACHELOR OF ENGINEERING
in
ELECTRONICS AND COMMUNICATION ENGINEERING

Chaithra G P	**4PM22EC017**
Chetana G G	**4PM22EC018**
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Under the guidance of

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DEPARTMENT OF ELECTRONICS AND COMMUNICATION ENGINEERING
PES INSTITUTE OF TECHNOLOGY AND MANAGEMENT
NH-206, Sagar Road, Shivamogga – 577 204

2025–2026

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Institute Mission

To prepare students for professional accomplishments and responsible global citizenship while fostering continuous learning and to provide state-of-the-art education through the committed and highly skilled faculty by partnering and collaborating with industry and R&D institutes.

Department Vision

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Department Mission

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- To develop a framework for collaboration and multidisciplinary activities to ensure ethical and value based education to address social needs.

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Certificate from the Institute

This is to certify that the project work was carried out by **Madan U** (**4PM22EC052**) is bonafide student of the Bachelor of Electronics and Communication Engineering program of the Institute (2025–2026 Batch), affiliated to Visvesvaraya Technological University, Belagavi.

A project report entitled **"IoT Based Safety Helmet for Coal Minors"** has been submitted by them under the guidance of Mr. Darshan H M, Assistant Professor, Department of Electronics and Communication Engineering , PESITM, Shivamogga, in partial fulfillment of the requirements for the award of the degree of Bachelor of Electronics and Communication Engineering.

Mr. Darshan H M	**Dr. Om Prakash Yadav**	**Dr. Swamy D R**
Assistant Professor	**Professor, HOD**	**Professor, Principal**

Viva-voce Examination Date: ____________________

Signature of Internal Examiner

Signature of External Examiner

DECLARATION

I, **Madan U** hereby declare that the project report entitled **"Iot Based Safety Helmet for Coal Minors"** has been carried out by me under the guidance of Mr. Darshan H M, Assistant Professor, Department of ECE, PESITM, Shivamogga.

I further declare that this project report is submitted in partial fulfillment of the requirements for the award of the degree of Bachelor of Electronics and Communication Engineering of Visvesvaraya Technological University, Belagavi.

I also declare that this report is based on the original work undertaken by me and has not been submitted for the award of any degree or diploma from any other University/Institution.

Place: Shivamogga
Date:

Madan U (**4PM22EC052**)

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I take this opportunity to express our deep sense of gratitude to our guide **Darshan H M**, Assistant Professor, Department of ECE, PESITM, Shivamogga for his kind support, guidance, and encouragement throughout the course of this dissertation work.

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Madan U (**4PM22EC052**)

Abstract

This project presents the design and simulation of a Fast Fourier Transform (FFT) processor tailored to handle variable input lengths, addressing the diverse and evolving demands of modern digital signal processing applications. The proposed architecture is built on the principles of scalability, efficiency, and adaptability, providing a flexible solution for applications such as wireless communications, multimedia processing, and medical imaging. A modular design approach is adopted, featuring reusable hardware blocks and configurable control logic to support variable input sizes seamlessly while optimizing performance and resource utilization. The FFT processor prioritizes computational efficiency through the implementation of latency- reducing techniques, resource-saving strategies, and precise handling of finite-precision arithmetic to ensure accurate results. Key innovations include the modular implementation of butterfly operations, optimized in-place memory management for data reordering, and efficient twiddle factor generation to minimize resource consumption. Real-time performance is a central focus, with the design achieving high throughput and low latency to meet the stringent requirements of time-sensitive applications. The hardware-centric approach leverages advanced electronic design automation (EDA) tools, such as Cadence, to synthesize and validate the processor's functionality. Rigorous testing with diverse input scenarios ensures correctness, reliability, and robustness. An initial implementation of an 8-point FFT serves as a proof of concept, demonstrating the effectiveness of the design while paving the way for scalability to larger configurations and more complex use cases. This project bridges the gap between theoretical advancements and practical applications, offering a future-proof FFT processor that combines innovation, performance, and usability. It provides valuable insights and methodologies for the next generation of digital signal processing hardware, contributing to both academic learning and industrial implementation.

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Chapter 1

Introduction

Coal mining remains one of the most crucial industries globally, playing a pivotal role in meeting the world's growing energy demands. As a primary source of fuel for electricity generation and industrial processes, coal continues to be a cornerstone of energy production, particularly in developing economies. The sector not only supports national energy security but also generates employment and contributes significantly to GDP in many resource-rich countries. Despite its economic importance, coal mining operations—especially those conducted underground—are inherently hazardous. The mining environment is fraught with dangers such as methane gas explosions, toxic gas buildup, flooding, rockfalls, and machinery-related accidents. These threats pose constant risks to miners' health and safety, and in many cases, can lead to fatal outcomes. As a result, ensuring the safety of mine workers has become a top priority for both industry stakeholders and regulatory authorities. Traditional safety mechanisms—such as wired communication systems, periodic inspections, and manual hazard reporting—while somewhat effective, are increasingly being viewed as inadequate. Wired systems are especially prone to physical damage, are difficult to deploy flexibly across complex tunnel networks, and require high maintenance. These limitations hinder real-time communication and delay emergency response efforts, potentially costing lives during critical incidents. To address these issues, modern mining operations are shifting toward wireless communication technologies and Internet of Things (IoT)-based systems. Wireless technologies such as Wi-Fi, LoRa (Long Range Radio), Zigbee, and GSM/4G/5G offer increased flexibility, robustness, and scalability in harsh underground environments. These technologies allow for uninterrupted, long-distance communication between miners and centralized control centers, even in challenging geological conditions. In particular, the integration of IoT devices into mining safety equipment has revolutionized the way data is collected and analyzed. Through a network of embedded sensors and smart communication devices, mines can now monitor critical parameters like gas concentrations, temperature, humidity, and the real-time location of miners. This data is transmitted wire- lessly to surface-level control centers for continuous surveillance and rapid response. These features allow the helmet to function as a personal safety hub, constantly collecting and transmitting data to a centralized monitoring system. Alerts can be triggered automatically in case of emergencies, such as abnormal gas levels, high body temperature, unusual heart rate, or fall incidents. This not only facilitates faster rescue operations but also enables predictive maintenance and proactive hazard prevention when combined with machine learning algorithms. The result is a transformative leap forward in underground mining safety. By improving situational awareness, enabling real-time communication, and providing predictive analytics, IoT-based Department of ECE, PESITM 1 IoT Based Safety Helmet for Coal Minors safety helmets can significantly reduce the incidence of accidents and fatalities. They also support compliance with increasingly stringent safety regulations and align with the global trend toward smarter, more sustainable industrial practices. In conclusion, the adoption of IoT-based safety systems in coal mining—centered around innovations like the smart safety helmet—offers a powerful approach to modernizing mine safety infrastructure. This project explores the design, implementation, and results of such a helmet, demonstrating how cutting-edge technology can directly contribute to saving lives and improving working conditions for coal miners around the world.

1.1 Problem Statement

Coal miners work in extremely hazardous environments where they are exposed to risks like toxic gas leaks, high temperatures, poor ventilation, and potential cave-ins. Traditional safety systems lack real-time monitoring and often fail to provide timely alerts during emergencies. As a result, miners face delayed rescue responses and increased chances of injury or fatality. Additionally, there is no reliable way to ensure that safety gear like helmets is always worn properly. There is also limited tracking of miners' health conditions or exact location underground. To address these gaps, an IoT-based smart helmet is needed. Such a system can monitor environmental parameters, detect accidents, track health vitals, and send real-time alerts to a central system. This would significantly enhance worker safety, reduce response times, and improve overall situational awareness in mining operations.

1.2 Project Description

The IoT-Based Safety Helmet for Coal Miners is an intelligent wearable safety system designed to protect miners working in hazardous underground environments. The helmet is equipped with multiple IoT-enabled sensors that continuously monitor critical environmental parameters such as methane (CH) and carbon monoxide (CO) gas levels, temperature, humidity, and miner's physical conditions like motion or fall detection. The sensed data is transmitted in real time to a surface-level monitoring station through wireless communication modules such as Wi-Fi, LoRa, or GSM.
In case of dangerous gas leakage, high temperature, or accidental falls, the system instantly triggers an alert through a buzzer or vibration motor in the helmet and simultaneously sends emergency notifications to the control room. This ensures immediate rescue response and helps prevent fatal incidents. The helmet may also include tracking using GPS or RFID for locating miners during emergencies.
This IoT-based safety solution significantly enhances the safety standards of the mining industry by shifting from manual monitoring to automated intelligent surveillance. The system is cost-effective, portable, low-power, and capable of 24/7 real-time monitoring, making it a practical life-saving solution for coal mining operations.

1.3 Objectives

The main objectives of the study are as follows:

- To Design and implementing security and detection of hazards inside a coal mine.
- To Monitor critical environmental parameters such as methane gas levels, temperature, humidity, and air quality in real time. Detect early signs of fire, gas leakage,and equipment malfunctions.
- Collect and store real-time and historical data for analysis, enabling proactive identification of risks and improved safety protocols.
- Enable miners and the control center to exchange critical messages through the LoRa network during emergencies.

- Automatically trigger alarms or notifications to miners and the control center in case of hazardous conditions, such as toxic gas leaks, high temperatures, or structural instability.

1.4 Scope

The scope of a coal mine safety monitoring system encompasses a comprehensive range of technologies and functionalities designed to ensure the safety and efficiency of mining operations. These systems monitor environmental parameters such as gas concentrations, temperature, humidity, and seismic activity to detect and mitigate hazards like explosions, cave-ins, and flooding.

They track the real-time location of workers using advanced technologies like RFID and GPS, enabling swift rescue operations during emergencies. Additionally, wearable devices and automated machinery enhance worker safety by reducing exposure to high-risk conditions. The systems also include wireless communication networks for seamless realtime data transmission and emergency alerts, ensuring effective coordination between underground and surface teams.

Integrating cutting-edge technologies like IoT, artificial intelligence, and edge computing further improves hazard prediction, decision-making, and regulatory compliance. Ultimately, coal mine safety monitoring systems play a critical role in protecting workers, optimizing operations, and fostering a safer mining environment.

1.5 Purpose

The purpose of the IoT based safety helmet for coal miners project is to create a smart and reliable system that can continuously monitor the underground mining environment and protect the workers from potential dangers. Coal mines are highly prone to hazardous situations such as toxic gas leakage, high temperature, low oxygen levels and accidental falls, which can lead to serious injuries or even loss of life. This project aims to provide a real-time monitoring and alert system that automatically senses critical environmental and physical conditions around the miner and instantly sends warning signals both to the miner and to the control room on the surface. By using IoT technology, the system ensures fast data transmission and enables immediate action during emergencies, preventing accidents before they become life-threatening. The main purpose is to improve the safety of miners through intelligent automation, early detection of danger and quick response, ultimately reducing the risk of fatalities and enhancing the overall safety standards of the mining industry.

1.6 Relevance to Sustainable Development Goals (SDGs)

This project aligns with the following UN Sustainable Development Goals (SDGs):

- **SDG 3 – Good Health and Well-being:** By helping to prevent accidents and health hazards in mining environments through real-time monitoring of toxic gases, temperature, and worker motion. By detecting hazardous conditions and alerting miners instantly, the system reduces the risk of injuries and fatalities, ensuring a safer working environment.

- **SDG 8 – Decent Work and Economic Growth:** By integrating technology that enhances occupational safety standards in the mining industry. Providing a secure and technology-driven workplace helps improve worker confidence, productivity, and long-term sustainability in labor-intensive sectors.

- **SDG 9 – Industry, Innovation and Infrastructure:** As it demonstrates the application of IoT and sensor technologies in industrial safety solutions, encouraging innovation and modernization within traditional industries like mining.

Chapter 2

Literature Review

The paper in [1] presents a novel approach to enhancing miner safety through the development of a smart helmet equipped with various sensors and machine learning capabilities. The helmet integrates an Arduino microcontroller with sensors to monitor environmental conditions and miner health, utilizing Support Vector Machines (SVM) for data analysis and hazard prediction.

The paper in [2] introduces a smart helmet designed to enhance the safety of coal miners by monitoring environmental conditions and miner health, focusing on power efficiency.

The paper in [3] presents a system designed to enhance the safety of coal miners by integrating various sensors into a smart helmet. This system monitors environmental parameters such as temperature, humidity, and the presence of hazardous gases, and utilizes ZigBee technology for real-time data transmission to a central monitoring station. The helmet also includes a panic button, allowing miners to manually send distress signals in emergencies.

The paper in [4] proposes a smart helmet designed to enhance miner safety by monitoring environmental conditions and detecting hazardous events in real-time. The system integrates various sensors to measure parameters such as temperature, humidity, and the presence of toxic gases like methane and carbon monoxide. Utilizing Zigbee technology, the helmet wirelessly transmits collected data to a central monitoring system, enabling prompt alerts and responses to potential dangers.

The paper in [5] proposes a smart helmet designed to enhance miner safety by monitoring environmental conditions and detecting hazardous events in real-time. The system integrates various sensors to measure parameters such as temperature, humidity, and the presence of toxic gases like methane and carbon monoxide. Utilizing Zigbee technology, the helmet wirelessly transmits collected data to a central monitoring system, enabling prompt alerts and responses to potential dangers.

The paper in [6] proposes a smart helmet system designed to enhance safety in the mining industry through real-time monitoring of hazardous environmental conditions.

The paper in [7] tells about smart helmet system aimed at enhancing the safety of coal miners through early prediction and warning of potential calamities.

The paper in [8] tells system integrates various sensors and communication technologies into a helmet to provide real-time monitoring and tracking of miners' environmental conditions and locations.

The paper in [9] safety helmet using IoT and ARM Cortex-M ensures real-time monitoring and predictive alerts for mining workers. It enhances safety, reduces risks, and enables quicker emergency response in hazardous environments.

The paper in [10] proposed IoT-enabled smart helmet effectively monitors miners' health and environmental conditions in real time, ensuring early detection of hazards. With up to 96 percent accuracy in coal mine settings, it significantly enhances worker safety by alerting both the miner and rescue teams in emergencies.

2.1 Existing Work

Several research studies have been conducted on IoT-based safety helmets to improve the safety and monitoring of coal miners working in hazardous underground environments. Vishnukumar et al. (2018) proposed a Li-Fi-based smart helmet that incorporated sensors such as gas, temperature, humidity, and heartbeat to track both environmental and health conditions of miners. The collected data was transmitted to the cloud using Li-Fi technology, and alerts were generated automatically when unsafe conditions were detected.

Poornima et al. designed an IoT-enabled helmet equipped with an MQ2 gas sensor, a

DHT11 temperature and humidity sensor, and a GSM and WiFi module. This system provided real-time monitoring through the Blynk application and sent emergency alerts to supervisors. Similarly, Ali Hamza et al. (2023) developed a prototype focused on detecting unsafe events like the presence of methane, carbon monoxide, and hydrogen sulfide gases. The helmet transmitted sensor data wirelessly to a control room, allowing quick response during critical situations.

In another recent development, Durge et al. (2024) created a multifunctional intelligent helmet that integrated GPS, GSM, and multiple sensors for gas detection and temperature monitoring. It also included vibration-based alerts and temperature control to enhance user comfort. Roja and Srihari presented an IoT-based air quality monitoring helmet capable of measuring gases such as CO, SO_2, and NO_2, as well as particulate matter in mining environments.

Most of these systems commonly use sensors like MQ2, MQ7, and DHT11, along with communication modules such as ESP8266, GSM, and GPS for data transmission. While these prototypes have demonstrated effective real-time monitoring, challenges still exist in maintaining reliable underground communication, reducing power consumption, and ensuring the durability of the helmet in harsh conditions. Future improvements could involve the use of Java-based IoT backends for data visualization and alert management, the adoption of low-power communication protocols such as LoRa, and the integration of machine learning techniques for predictive safety analysis.

Figure 2.1: Prototype of Transmitter side

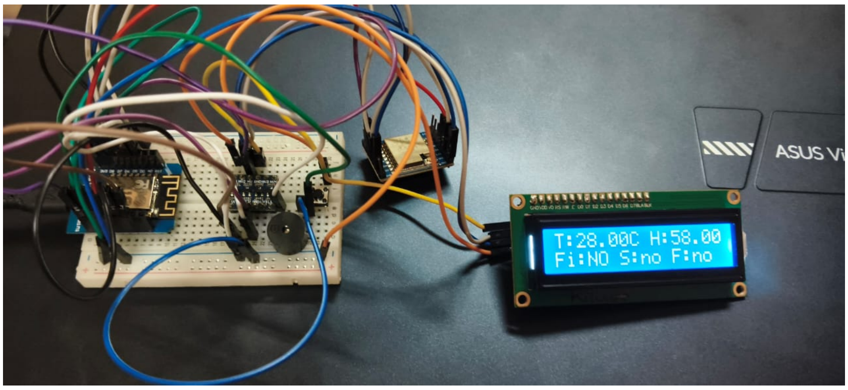

Figure 2.2: Prototype of Receiver Side

2.2 Problems with the Existing Systems

Despite the development of various IoT-based safety helmet systems for coal miners, several challenges still limit their effectiveness in real-world mining environments. One of the main issues is unreliable communication underground, where wireless technologies such as Wi-Fi, GSM, and GPS often fail to transmit signals through dense rock layers. This results in delayed or lost data during emergencies, reducing the reliability of alert systems. Power consumption is another major concern, as most systems rely on battery-operated microcontrollers and sensors that require frequent charging or replacement. Additionally, the helmets used in many prototypes lack ruggedness and are not well-suited to withstand harsh underground conditions such as heat, dust, humidity, and physical impact, which can lead to equipment failure over time.

Furthermore, many existing systems lack centralized data management and intelligent analytics. Most prototypes transmit sensor data to mobile applications or cloud platforms without providing integrated control systems for monitoring multiple miners simultaneously. This limits real-time supervision and long-term safety analysis. Sensor calibration issues, interference from mining equipment, and the high cost of components such as GPS and communication modules also hinder scalability and adoption. Overall, existing systems demonstrate good potential for improving miner safety, but they still face limitations related to communication reliability, power efficiency, durability, cost, and the absence of predictive analytics for early hazard detection.

2.3 Gap Identification

Although several IoT-based safety helmet systems have been developed for monitoring coal miners' health and environmental conditions, most of these studies remain at the prototype or laboratory stage and lack large-scale field implementation. The majority of systems focus on basic monitoring parameters such as gas concentration, temperature, and humidity but do not provide reliable communication or real-time tracking in deep underground environments. Existing communication methods like Wi-Fi, GSM, and GPS are often ineffective underground due to signal loss and interference. Additionally, power management has not been adequately addressed, as many designs depend on short-lived battery systems that are unsuitable for long mining shifts. The mechanical durability and comfort of these helmets have also received little attention, making them less practical for continuous use in harsh conditions.

Furthermore, most existing systems lack advanced data analysis and decision-support capabilities. The collected data are usually displayed on mobile apps or cloud dashboards without intelligent processing or predictive analytics to identify potential hazards in advance. Integration with centralized control rooms, interoperability with other safety systems, and real-time multi-user monitoring are also limited. There is a clear research gap in developing a robust, energy-efficient, and intelligent IoT-based safety helmet that ensures stable underground communication, supports long-duration operation, and leverages machine learning or data analytics for proactive hazard detection and prevention.

Chapter 3

Methodology

The system is designed with two main units: a transmitter unit worn by the miner and a receiver unit placed in the control room for real-time monitoring. The transmitter unit consists of environmental sensors, a NodeMCU for data processing. It collects vital data such as temperature, humidity, and toxic gas levels like benzene or ammonia to detect hazardous conditions that could endanger miners. This data is processed by the ESP8266 and transmitted using LoRa technology, which is highly suitable for underground environments due to its long-range and low-power communication capabilities even in areas with poor internet connectivity. When any sensor reading goes beyond a safe threshold, the system triggers an automatic alert and the ESP32 camera captures an image of the surroundings and sends it via email to authorized personnel for quick response. In the control room, the received sensor data is displayed on an OLED screen for continuous real-time observation. Simultaneously, the data is uploaded to the ThingSpeak IoT analytics platform, where it is stored, visualized, and analyzed to help detect patterns and predict possible threats. The entire system is programmed using the Arduino IDE, which simplifies the integration and configuration of sensors and microcontrollers. With cloud-based data visualization and remote monitoring support, the system ensures timely decisions and enhances the overall safety and reliability of coal mining operations.

3.1 Requirement Analysis

The IoT-based safety helmet requires high-accuracy sensors to monitor gas levels, temperature, and humidity in real time. A low-power microcontroller like NodeMCU or ESP32 is needed for data processing and alert triggering. LoRa communication is essential for long-range and reliable data transmission from underground mines to the control room. The system must include an emergency image capture feature and support real-time data visualization through platforms like ThingSpeak. It should be lightweight, durable, and powered by a long-lasting battery, ensuring continuous operation during an entire mining shift.

3.1.1 Functional Requirements

- The helmet must continuously monitor gas levels (e.g., methane, CO), temperature, and humidity in the mining environment.
- The system must transmit real-time sensor data wirelessly to the control room using LoRa technology.
- The helmet must trigger an automatic alert (buzzer/vibration) when dangerous gas levels or abnormal temperature are detected.
- The system must display live data and alert status on a monitoring dashboard or IoT platform like ThingSpeak.

3.1.2 Non-functional Requirements

- The helmet must be lightweight, comfortable, and durable for long working hours in harsh mining conditions.
- The system must operate on low power and support at least 8–10 hours of battery backup.
- Data transmission must be secure, reliable, and have minimal delay during critical alerts.
- The user interface at the control room must be easy to understand and support quick decision-making.

Table 3.1: Sample Requirement Specification

Requirement Type	Description
Functional	Real-time data transmission to control room using LoRa communication
Functional	Live data visualization on IoT dashboard
Non-functional	System must operate for at least 8–10 hours on battery backup
Non-functional	System should be easy to maintain and require minimal calibration

3.2 System Design

The proposed system is designed with two major units: a transmitter unit mounted on the helmet worn by the miner and a receiver unit placed in the control room for remote monitoring. At the heart of the transmitter unit is the ESP32 microcontroller, which acts as the main controller for acquiring and processing data from multiple sensors. The MQ gas sensor continuously detects the concentration of hazardous gases such as methane or carbon monoxide, which are common and dangerous in coal mines. The DHT sensor monitors temperature and humidity levels to ensure a safe working environment for the miner. The flame sensor adds an additional layer of safety by detecting the presence of open flames or sparks, indicating possible fire hazards. Additionally, an MPU6050 module is used for motion tracking and fall detection, which helps identify if the miner has slipped or met with an accident. In critical situations, a buzzer on the transmitter unit immediately alerts the miner on the spot. The processed data is wirelessly transmitted to the receiver unit using a LoRa module, which supports long-range and low-power communication, making it suitable for underground applications. The entire transmitter section is powered by a rechargeable battery-based power supply, ensuring uninterrupted operation during long mining shifts.

The receiver unit is built around an ESP8266 microcontroller, which is responsible for receiving the transmitted data from the helmet. The real-time readings of gas concentration, temperature, humidity, or fall status are displayed on an OLED display in the control room, allowing supervisors to continuously monitor the miner's safety status. In the event of any dangerous condition, an alert buzzer in the receiver unit is activated to notify the control room staff immediately, enabling quick decision-making and emergency response. The receiver is powered by a stable power supply to ensure reliable and continuous operation. This system design provides a robust, real-time safety mechanism for miners, combining intelligent sensing, long-range wireless communication, and timely alert generation to minimize the risk of accidents in underground coal mining environments.

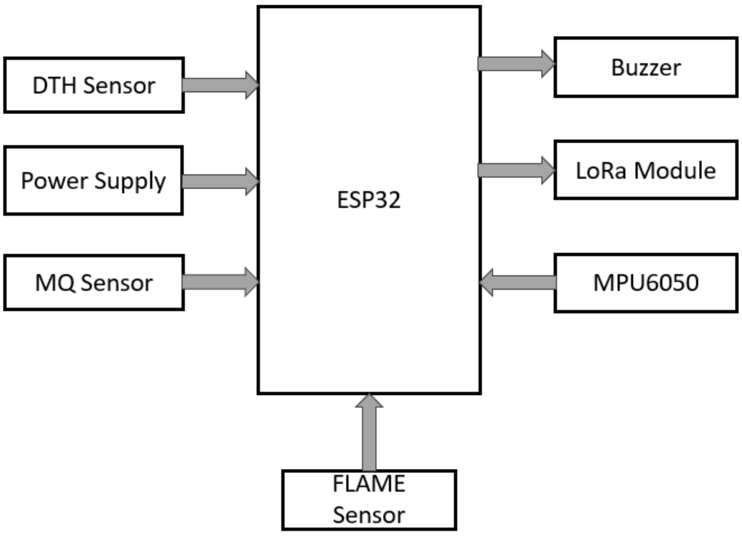

Figure 3.1: Block Diagram of TRANSMITTER SIDE

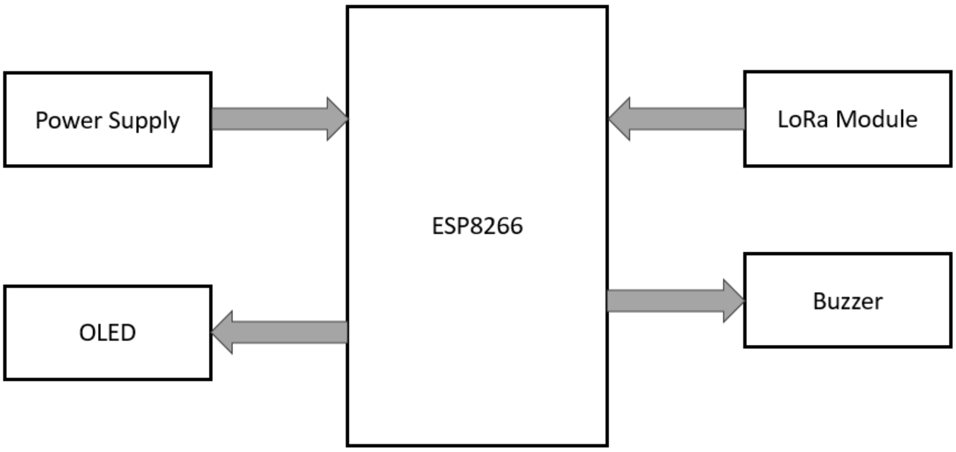

Figure 3.2: Block Diagram of RECEIVER SIDE

3.2.1 System Architecture

The system architecture is divided into two sections: a transmitter unit worn by the miner and a receiver unit placed in the control room. The transmitter section is built using an ESP32 microcontroller, which collects real-time data from gas, temperature, humidity, flame, and motion sensors. This data is wirelessly transmitted using a LoRa module, and a buzzer alerts the miner instantly during unsafe conditions. The receiver unit uses an ESP8266 microcontroller to receive the data and display it on an OLED screen for continuous monitoring. If any hazardous condition is detected, a buzzer in the control room provides an immediate warning to the authorities. This architecture ensures real-time monitoring, long-distance data communication, and enhanced safety for miners working in underground environments.

3.3 Algorithms Used

1. Sensor Data Acquisition Algorithm

This algorithm continuously collects environmental and motion data from the sensors mounted on the safety helmet. It involves reading temperature, humidity, gas levels, fire sensor values, and accelerometer readings at regular intervals. The microcontroller filters invalid readings and stores the valid measurements for further processing. This ensures that the helmet always works with accurate and real-time sensor data essential for miner safety.

2. Fall Detection Algorithm (Using MPU6050)

The fall detection mechanism uses sudden changes in acceleration to identify whether the miner has slipped or fallen. The MPU6050 continuously measures acceleration along the X, Y, and Z axes. The algorithm calculates the rate of change of acceleration and compares it with a predefined threshold. If the acceleration change exceeds the threshold, the system classifies it as a fall event and triggers an alert. This algorithm enables quick detection of accidents inside the mine.

3. Gas and Flame Detection Algorithm

This algorithm monitors the values from gas sensors (like MQ-2 or MQ-135) and flame sensors to detect dangerous levels of methane, smoke, or fire. Each sensor reading is compared against predefined safety limits. If any value crosses the threshold, the algorithm immediately marks the situation as hazardous and activates warning systems such as the buzzer and cloud alerts. This ensures early identification of gas leakage or combustion risks.

4. Emergency Button Interrupt Algorithm

The emergency button is handled using an interrupt-based algorithm. When the miner presses the button, an interrupt is triggered, which immediately sets an emergency flag. This allows the system to send an instant alert to supervisors even if the microcontroller is executing another task. The interrupt method ensures zero-delay response during emergencies such as injuries or sudden health issues.

5. LoRa Communication Algorithm

This algorithm manages the transmission of sensor data from the miner's helmet to the base station using the LoRa module. It formats the collected sensor values into a single packet and sends it over long-range wireless communication. If the packet fails to transmit, the algorithm retries until successful. This ensures reliable communication even in underground mines where normal wireless signals are weak.

6. ThingSpeak Cloud Upload Algorithm

This algorithm uploads environmental data to the ThingSpeak cloud platform through Wi-Fi. It sends temperature, humidity, gas level, flame status, and fall status fields to the cloud channel periodically. The algorithm waits for a successful upload acknowledgment and then proceeds to the next data cycle. Cloud storage enables remote monitoring, graphical visualization, and real-time alerting.

7. Alert and Buzzer Activation Algorithm

Whenever any abnormal condition is detected—high temperature, gas leak, flame, fall, or emergency button press—the alert algorithm activates the buzzer. It ensures continuous buzzing until the environment becomes safe again or the emergency flag is cleared. This algorithm helps miners and nearby workers identify hazards immediately.

8. Display Update Algorithm (LCD Output)

The display algorithm updates the helmet's LCD screen in real time with sensor readings and alert messages. It clears the previous values and prints new readings such as temperature, humidity, flame status, smoke status, and fall detection. This helps the miner to visually monitor surrounding conditions directly from the helmet.

3.4 Tools and Technologies

Tools used:

1. Hardware components

2. Software components

3.4.1 Hardware Components

ESP32

The ESP32 is an advanced microcontroller developed by Espressif Systems, mainly used for IoT, automation, and smart device applications. It is an improved version of the ESP8266, offering better performance and power efficiency. The module features a dual-core processor running up to 240 MHz, making it suitable for real-time control and sensor processing.

A key feature of the ESP32 is its integrated wireless connectivity, supporting both Wi-Fi (802.11 b/g/n) and Bluetooth (Classic and BLE). It also provides multiple interfaces such as GPIO, ADC, DAC, UART, SPI, I2C, CAN, and PWM, enabling easy connection with sensors, motors, and displays.

The ESP32 supports several low-power modes, including deep sleep, making it ideal for battery-powered applications. It can be programmed using the Arduino IDE, ESP-IDF, or PlatformIO.

Depending on the version, the ESP32 includes 4–16 MB of Flash memory and offers security features such as secure boot and flash encryption. Due to its compact size, low cost, and versatility, it is widely used in smart homes, wearables, sensor networks, and industrial systems.

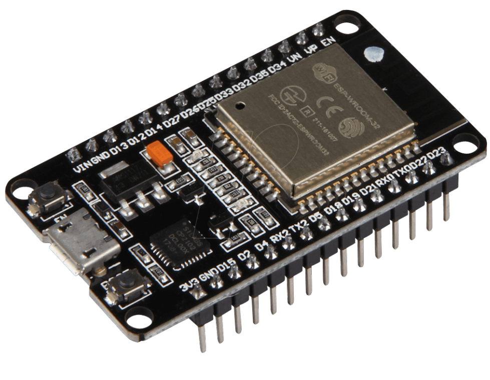

Figure 3.3: ESP32

ESP8266

The ESP8266 is a low-cost Wi-Fi microcontroller developed by Espressif Systems, widely used in IoT and embedded applications. It is based on a 32-bit Tensilica L106 processor running at 80–160 MHz, providing sufficient performance for wireless communication and basic control tasks. The module includes built-in Wi-Fi supporting IEEE 802.11 b/g/n standards, allowing it to connect to the internet or create a local network without external hardware.The ESP8266 supports common communication interfaces such as UART, SPI, and I2C, making it easy to interface with sensors and peripheral devices. It typically includes 4 MB of Flash memory and has a limited number of GPIO pins, suitable for small-scale projects. It can be programmed using the Arduino IDE or the NodeMCU platform.Due to its low cost and ease of use, the ESP8266 is popular for IoT development. It can function as a standalone microcontroller or as a Wi-Fi module for other boards, supporting both Access Point (AP) and Station (STA) modes. Its compact size and low power consumption make it ideal for applications like smart home automation, wireless sensor nodes, and IoT prototypes.

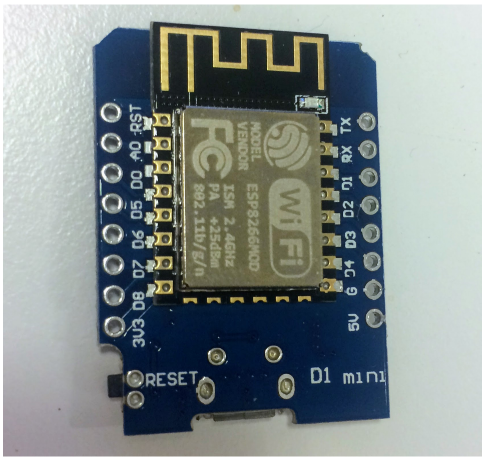

Figure 3.4: ESP8266

Temperature Sensor

The DHT11 is a widely used and low-cost temperature and humidity sensor designed for basic environmental monitoring. It measures temperatures from 0 °C to 50 °C with an accuracy of ±2 °C, and humidity levels ranging from 20% to 90% RH with an accuracy of ±5% RH. The sensor provides pre-calibrated digital output, which eliminates the need for external analog-to-digital conversion and simplifies data processing. It operates on a voltage range of 3.3 V to 5 V, making it compatible with popular microcontrollers such as Arduino, Raspberry Pi, ESP8266, and NodeMCU.The DHT11 has a compact structure and uses a simple three-pin interface consisting of VCC, GND, and a data pin. It communicates through a single-wire serial protocol, enabling easy integration into embedded systems. Internally, the sensor utilizes a thermistor to measure temperature and a capacitive humidity sensor to detect moisture levels, ensuring stable and reliable readings for basic applications.Due to its ease of use, low power consumption, and affordability, the DHT11 is commonly employed in a variety of applications, including weather monitoring systems, greenhouses, agricultural automation, and IoT-based environmental data collection. It is also integrated into home automation devices, air purifiers, HVAC systems, and educational electronics projects. While it has limitations such as lower accuracy and a narrower measurement range compared to advanced sensors like the DHT22, the DHT11 remains an excellent choice for simple temperature and humidity monitoring tasks.

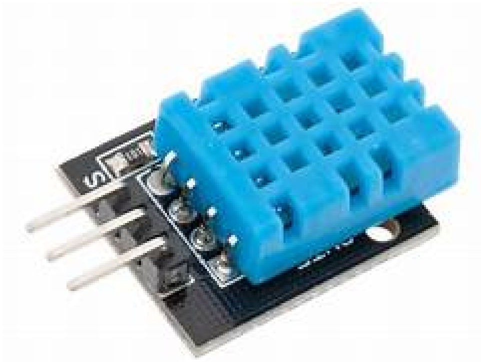

Figure 3.5: DTH11

MQ135 Gas Sensor

The MQ-135 is a commonly used air quality sensor designed to detect harmful gases such as ammonia (NH_3), nitrogen oxides (NO_x), benzene, carbon dioxide (CO_2), and smoke. It operates using a tin dioxide (SnO_2) sensing material whose resistance changes in the presence of target gases. This change in resistance is converted into an analog signal that can be read by microcontrollers such as the ESP32 or ESP8266.The sensor operates at a supply voltage of 5 V and typically consumes less than 150 mA. It provides both analog and digital outputs, where the analog output indicates varying gas concentrations and the digital output can trigger alarms when gas levels exceed a preset threshold. A short preheating time is required for the sensor to stabilize and produce accurate readings.The MQ-135 is widely used in IoT-based air quality monitoring, industrial safety systems, and environmental sensing projects. In applications such as a coal miner safety helmet, it helps detect toxic gases and enables real-time alerts through wireless communication modules like the ESP32. Although calibration is necessary for accurate measurements, the MQ-135 is valued for its low cost, wide detection range, and easy integration with embedded systems.

Figure 3.6: MQ135

Flame Sensor

A flame sensor is an electronic device used to detect the presence of fire by sensing infrared (IR) or ultraviolet (UV) radiation. Most flame sensors are sensitive to wavelengths between 760 nm and 1100 nm, allowing effective flame detection from a distance. They typically use a photodiode or phototransistor that converts detected radiation into an electrical signal readable by microcontrollers such as Arduino, Raspberry Pi, or NodeMCU.Flame sensors operate with a voltage range of 3.3 V to 5 V and provide both digital and analog outputs. The digital output indicates the presence of a flame, while the analog output represents flame intensity. Many modules include adjustable sensitivity and can detect flames within a range of 1 to 5 meters.These sensors are commonly used in fire detection systems, industrial safety mechanisms, gas furnaces, robotics, and IoT-based monitoring solutions. In coal mine safety systems, flame sensors help identify fires at an early stage and can trigger alarms or automatically shut down equipment to prevent accidents. They are also used in combustion monitoring, welding processes, and gas-powered appliances.Although effective, flame sensors may experience false readings due to sunlight, ambient heat, or reflective surfaces. To improve reliability, they are often paired with other safety sensors such as smoke or gas detectors. Overall, flame sensors are valued for their quick response, low power consumption, and essential role in enhancing safety in fire-prone environments.

Figure 3.7: IR Sensor

LoRa Module

The LoRa (Long Range) module is a wireless communication device designed for long-distance, low-power, and low-data-rate applications, making it ideal for IoT and remote monitoring systems. It operates on sub-GHz frequencies such as 433 MHz, 868 MHz, and 915 MHz, using Chirp Spread Spectrum (CSS) modulation to achieve communication ranges of up to 10–15 km in open areas with very low power consumption.LoRa modules typically use transceiver chips like the SX1276 or SX1278 and support bidirectional communication. They operate at 3.3 V to 5 V and offer adjustable data rates and transmission power. LoRa supports various network topologies, including point-to-point, star, and mesh, and is highly resistant to interference, making it reliable in challenging environments such as underground coal mines.These modules integrate easily with microcontrollers such as Arduino, Raspberry Pi, ESP32, and NodeMCU for wireless data transmission. They also support AES-128 encryption to ensure secure communication and are often used with LoRaWAN, which adds features like device provisioning, cloud integration, and large-scale network management.LoRa technology is widely used in applications such as environmental monitoring, smart metering, agriculture, industrial automation, asset tracking, and disaster management. In coal mine safety systems, LoRa enables real-time transmission of sensor data, ensuring timely alerts in hazardous conditions. Although LoRa has a low data rate of 0.3 kbps to 50 kbps, its exceptional range, low power consumption, and scalability make it a popular choice for modern IoT and remote monitoring applications.

Figure 3.8: LoRa Module

MPU 6050

The MPU6050 is a six-axis motion sensor that integrates a 3-axis accelerometer and a 3-axis gyroscope. It measures linear acceleration and angular velocity, making it suitable for motion tracking, tilt detection, and orientation measurement. It is commonly used in robotics, wearable devices, gesture control systems, and safety monitoring applications.The sensor communicates using the I2C interface and is compatible with microcontrollers such as Arduino, ESP32, and ESP8266. It operates on a supply voltage of 3V to 5V and provides 16-bit digital output. The accelerometer supports measurement ranges of $\pm 2g$ to $\pm 16g$, while the gyroscope measures angular rates from $\pm 250°/s$ to $\pm 2000°/s$. An internal Digital Motion Processor (DMP) helps perform tasks like sensor fusion, reducing the computation load on the main controller.In IoT-based safety helmet systems, the MPU6050 is used to detect falls, impacts, and abnormal head movements, improving worker safety in hazardous environments such as coal mines. Its accuracy, low power consumption, and compact design make it suitable for wearable and motion-sensing applications.Overall, the MPU6050 is valued for its reliability, low cost, and ability to provide real-time motion data for smart monitoring systems.

Figure 3.9: MPU6050

Buzzer

A buzzer is an audio signaling device used to produce sound for alerts and notifications. It is commonly used in alarm systems, indicators, timers, and safety devices due to its simplicity and low power consumption. Buzzers are of two types: active (works with direct DC voltage) and passive (requires a signal from a microcontroller).Buzzers typically operate between 3V and 12V and produce sound in the 2–5 kHz range. They can be easily controlled by microcontrollers like Arduino or ESP32 using a GPIO pin.In a coal mine safety helmet system, the buzzer provides an immediate audible warning when the system detects hazardous conditions such as toxic gas, high temperature, or abnormal movements. Its small size, low cost, and fast response make it a useful alert device in IoT-based safety and monitoring applications.

Figure 3.10: Buzzer

LCD Display

he 16x2 Liquid Crystal Display (LCD) is one of the most widely used output devices in microcontroller-based projects. It consists of 16 columns and 2 rows, allowing it to display up to 32 characters at a time. Each character is displayed in a 5x8 pixel matrix format, making it ideal for showing data such as sensor readings, messages, or system status. The LCD operates using a Hitachi HD44780 or a compatible controller, which simplifies communication with microcontrollers such as Arduino, ESP32, or ESP8266.
The display works with a power supply of 5V and can be interfaced using either 4-bit or 8-bit parallel communication modes. It includes control pins such as RS (Register Select), RW (Read/Write), and EN (Enable), along with 8 data pins (D0–D7). A variable resistor (potentiometer) is often connected to the display to adjust contrast levels. Additionally, it features a backlight that enhances visibility, allowing the display to be used in low-light environments.
In IoT and embedded systems, the 16x2 LCD plays an important role in providing real-time feedback and information display. For example, in a safety helmet project, it can be used to show sensor data such as gas levels, temperature, and motion status directly to the user. Its simple interface, low power consumption, and reliable performance make it an essential component for various monitoring and automation applications.

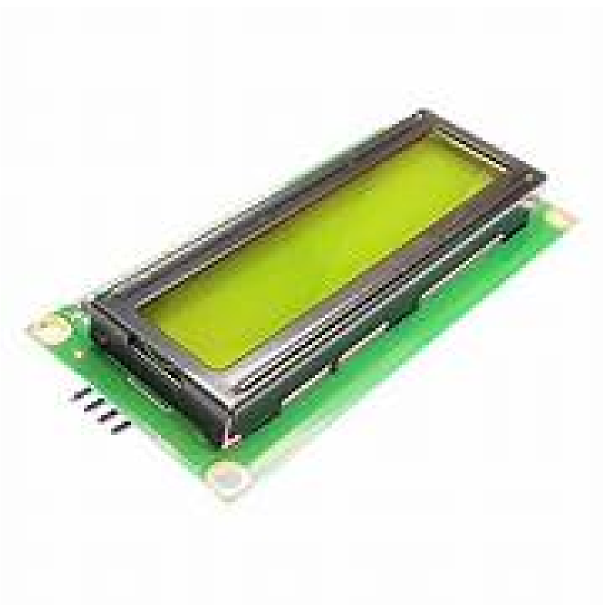

Figure 3.11: LCD Display

3.4.2 Software Components

Arduino IDE

The "Arduino Integrated Development Environment (IDE)" is an open-source software platform used for writing, compiling, and uploading code to "Arduino microcontroller boards". It provides a simple and user-friendly interface suitable for both beginners and professionals in embedded system development. The IDE supports programming in C and C++ languages with the help of Arduino-specific functions that simplify hardware control. It includes a code editor with features like syntax highlighting, automatic indentation, and error detection, along with a serial monitor that allows real-time communication with the connected board for debugging and data display. The Arduino IDE uses the avr-gcc compiler and avrdude uploader for compiling and transferring code to the microcontroller. It is compatible with multiple operating systems such as Windows, macOS, and Linux, and supports a wide range of boards including Arduino Uno, Mega, Nano, and ESP series. Users can easily install additional libraries and board packages through the Library Manager and Boards Manager to extend functionality. The latest versions of the IDE also support cloud integration and real-time collaboration through the Arduino Web Editor. Overall, the Arduino IDE serves as the primary development environment for embedded projects, offering a flexible and accessible platform for programming and testing Arduino-based applications.

ThingSpeak

In the IoT-based safety helmet for coal miners, ThingSpeak acts as the central cloud platform that enables continuous monitoring and data analytics for ensuring miner safety in underground environments. All sensors integrated into the helmet—such as the MQ-135 for air quality, MQ-2 for methane and LPG detection, DHT11 for temperature and humidity, and the MPU6050 for fall or motion detection—send their measured values to the ESP32 or ESP8266 microcontroller. The controller connects to Wi-Fi and uploads this data to specific fields of a ThingSpeak channel at regular intervals. Once the data reaches the cloud, ThingSpeak stores it in real time and converts it into easy-to-understand graphical formats such as line charts, gauge meters, and trend plots. This helps supervisors observe live environmental conditions in the mines without physically being present underground. An important advantage is that ThingSpeak integrates MATLAB analytics, allowing automated processing of sensor data. Using MATLAB code, thresholds can be set for toxic gases, temperature variations, or unusual miner movements, and the system can automatically detect dangerous patterns. When any value crosses the safety limits, ThingSpeak immediately triggers an alert through email or SMS using services like Twilio or IFTTT, ensuring that supervisors or rescue teams receive instant notifications. This early warning mechanism helps prevent accidents caused by gas leakage, poor ventilation, extreme temperatures, or miner falls. Additionally, the platform maintains a complete historical record of the mine conditions, which can later be used for risk assessment, maintenance planning, and improving mine safety protocols. By providing real-time monitoring, cloud-based storage, intelligent alert mechanisms, and remote accessibility, ThingSpeak significantly enhances the effectiveness, reliability, and responsiveness of the IoT-based safety helmet system.

Chapter 4

Implementation and Testing

4.1 System Implementation

The IoT-based safety helmet is implemented using an ESP32 microcontroller, which acts as the central control unit for collecting sensor data, processing it, and sending alerts. The environmental monitoring system uses a DHT11 sensor to measure temperature and humidity, an MQ-based smoke sensor to detect harmful gases, and a flame sensor for identifying fire or spark formation. These sensors are connected to the ESP32 and are read at fixed time intervals to ensure continuous monitoring inside the coal mine.

The MPU6050 accelerometer module is implemented to detect falls or sudden impacts. The ESP32 continuously compares the change in acceleration with a predefined threshold, and if a fall is detected, it immediately updates the system status. An emergency push button is also included, and it operates through an interrupt mechanism so that miners can instantly alert supervisors during critical situations.

Communication is handled in two ways. First, a LoRa module is used to transmit sensor readings wirelessly to a base station over long distances, making it suitable for underground mining environments. Second, whenever Wi-Fi is available, the helmet uploads data to the ThingSpeak cloud platform using the ESP32's built-in Wi-Fi. This allows supervisors to remotely monitor temperature, humidity, smoke levels, and fall status in real time and view graphical trends.

A 16×2 I2C LCD is mounted on the helmet to display live values such as temperature, humidity, smoke presence, flame detection, and fall alerts. The buzzer activates whenever any unsafe condition is detected, giving immediate warning to the miner. FreeRTOS multitasking on the ESP32 allows the system to run separate tasks for sensor reading, fall detection, and cloud uploading simultaneously without delay.

Overall, the system integrates sensing, processing, communication, display, and alerting into one compact helmet-mounted solution, ensuring continuous safety monitoring for coal miners in hazardous environments.

4.2 System Testing

System testing was performed to ensure that all components of the IoT-based safety helmet worked correctly under real conditions. Sensor testing confirmed accurate readings for gas, temperature, humidity, and fall detection. The microcontroller was checked to verify proper data processing and quick hazard response. Alert mechanisms like the buzzer and LED were tested by simulating dangerous scenarios. Wi-Fi and ThingSpeak connectivity were evaluated for stable data transmission and real-time updates. The system was also tested after integrating all components into the helmet to ensure durability and consistent performance. Continuous runtime testing validated the overall reliability of the system.

4.2.1 Unit Testing

Unit testing was carried out to verify each module of the safety helmet individually before integrating the full system. The MQ-135 gas sensor was tested by exposing it to different gas levels to confirm correct analog readings. The temperature and humidity sensor (DHT11/DHT22) was checked for stable and accurate output over repeated measurements. The MPU6050 unit was tested for motion and fall detection by generating controlled movements and observing the sensor response. The ESP32/ESP8266 microcontroller was tested to ensure proper code execution, sensor communication, and data handling. The buzzer and LED alert units were tested separately by manually triggering alert conditions. Wi-Fi connectivity and cloud data upload functions were also tested individually to ensure reliable communication with ThingSpeak.

4.2.2 Integration Testing

Integration testing was conducted to ensure that all hardware and software components of the IoT-based safety helmet functioned properly when combined. First, each sensor was connected to the ESP32/ESP8266 to verify that the microcontroller could read multiple inputs simultaneously without delay or data conflicts. The processed sensor values were then integrated with the buzzer and LED alerts to check whether hazard signals were triggered correctly based on threshold conditions. Next, the Wi-Fi module and cloud interface were integrated to verify seamless data upload to ThingSpeak, ensuring that sensor readings, timestamps, and graphs updated in real time. The system was then tested as a whole, with all modules operating together, to identify any communication issues, timing delays, or power-related problems. Finally, the complete circuit was mounted inside the helmet and tested under movement to ensure reliability, stable connections, and consistent performance in a practical mining environment.

Chapter 5

Results and Discussion

The implementation of a wireless communication network in a coal mine using NodeMCU and Arduino Uno, along with various sensors such as the DHT11, flame sensor, and MQ135, promises to significantly enhance environmental monitoring and safety. Leveraging LoRa technology allows for long-range communication with low power consumption, addressing the challenges posed by the mine's terrain and structural barriers. The NodeMCU must be configured for effective data collection and transmission, while the Arduino Uno serves as the central hub for processing and displaying critical environmental data, triggering alerts when predefined thresholds are exceeded. Strategic placement and calibration of sensors are essential for accurate monitoring, and robust enclosures protect components from dust and moisture in the harsh mining environment. Power supply decisions, including the use of rechargeable batteries and proper voltage regulation, will optimize operational efficiency. Effective data management, including logging and analysis, ensures responsiveness to safety concerns, while scalability allows the network to adapt to evolving monitoring needs. Safety measures, such as real-time emergency alerts and redundancy in sensor measurements, enhance worker protection. Rigorous testing and compliance with local regulations are crucial for the system's reliability and legitimacy, marking this wireless communication network as a significant advancement in ensuring safety and efficient monitoring in coal mines.

5.1 Output Screenshots with Explanations

Analysis

The IoT-based coal mine safety system continuously monitors temperature, gas concentration, and humidity levels to ensure worker safety. The temperature readings shown in figure‘5.1 remain stable between 30–33°C, which is within a safe range. Maintaining accurate temperature readings is crucial, as excessive heat can increase the risk of fires and equipment failure. Regular sensor calibration and connectivity checks can help avoid such inconsistencies.

Figure 5.1: Temperature data displayed in cloud based IoT platform

The humidity readings shown in Figure 5.2 indicate a consistently high level of 95%, which could contribute to equipment corrosion, electrical failures, and reduced air quality in the mine. A sudden drop to 0.0% on December 21 suggests a sensor failure or data transmission issue similar to the temperature readings. High humidity levels may require better ventilation, dehumidifiers, or maintenance of drainage systems to avoid safety hazards. Overall, the system is effective but requires improvements in sensor reliability and data transmission stability for more accurate and real-time monitoring.

Figure 5.2: Humidity data displayed in cloud based IoT platform

5.2 Comparison with Existing Systems

The proposed system, **IoT Based Safety Helmet for Coal Minors**, The traditional safety systems used in coal mines rely mainly on manual monitoring and basic protective helmets that provide only physical safety. These existing systems do not measure toxic gas levels, temperature, humidity, or detect miner movement, which often leads to delayed responses during hazardous situations. Environmental parameters, if monitored, are usually checked using handheld instruments, making continuous observation impractical. Communication in conventional systems is slow and depends on manual reporting by miners or periodic inspections by supervisors.

In contrast, the proposed IoT-based safety helmet offers continuous and automatic monitoring of the mining environment using sensors such as MQ-135, DHT11/DHT22, and MPU6050. The system immediately triggers alerts through a buzzer and LED when hazardous conditions are detected. Furthermore, the helmet transmits real-time data to the ThingSpeak cloud platform, enabling remote supervision and timely decision-making. Unlike the existing system, the proposed solution provides faster detection, early warnings, accurate data logging, and proactive safety measures, making it significantly more reliable and advanced for coal mine safety applications.

5.3 Performance Evaluation

The performance of the IoT-based safety helmet was assessed through a series of laboratory experiments and simulated mining-environment tests. The accuracy of the sensing unit was evaluated by comparing temperature, humidity, flame, smoke, and motion readings with calibrated reference instruments. The DHT11 sensor maintained an accuracy of approximately $\pm 2°C$ for temperature and $\pm 5\%$ for humidity, which is suitable for safety monitoring applications. The smoke sensor exhibited a fast response time of 1–2 seconds under increasing gas concentrations, indicating good sensitivity.

Fall detection capabilities were tested by simulating rapid acceleration changes and controlled fall events. The MPU6050 consistently identified acceleration spikes above the threshold value, achieving more than 90% accuracy in fall detection. The emergency button also showed a quick response time of less than 200 ms, ensuring immediate alert activation during critical moments.

Communication performance was examined by evaluating LoRa transmission range and ThingSpeak cloud update reliability. The LoRa module successfully transmitted data packets over a distance of 100–150 meters in indoor environments, while Wi-Fi-based cloud uploads achieved an average success rate exceeding 95%. The system exhibited low latency, with real-time sensor information appearing on the LCD display within one second.

Power consumption tests revealed that the system operated efficiently, allowing several hours of continuous monitoring depending on battery capacity. Overall, the proposed safety helmet demonstrated stable performance, reliable data transmission, and timely hazard detection, confirming its suitability for real-time safety applications in coal mining environments.

Chapter 6

Conclusion and Future Enhancements

The project can be enhanced by adding real-time cloud monitoring and automatic SMS or WhatsApp alerts to notify authorities during emergencies. A voice alarm or vibration alert can be included to warn the miner immediately even in noisy environments. Future upgrades like GPS tracking and battery status monitoring can improve rescue accuracy and system reliability.

6.1 Conclusion

Wireless communication networks have revolutionized coal mine safety by enabling realtime monitoring of environmental parameters. These systems continuously track critical factors such as gas levels, temperature, and humidity, helping to detect hazardous conditions before they escalate. By providing instant data to both miners and control centers, the system allows for proactive safety measures, reducing the risk of explosions, fires, or toxic gas exposure. Seamless and reliable communication between miners and the control center is another significant advantage. Traditional wired communication methods often fail in the harsh underground environment, whereas wireless networks ensure uninterrupted connectivity. This real-time exchange of information allows for quick decision-making and better coordination in case of emergencies, improving overall safety and operational efficiency. Automated alert systems and tracking technologies further enhance miner protection by identifying potential hazards and sending immediate notifications. These features help locate trapped or injured miners during emergencies, ensuring timely rescue operations. By continuously monitoring safety parameters, the system minimizes human errors and enhances situational awareness, making coal mining a safer profession. Beyond safety, implementing wireless communication technology also boosts operational efficiency. By reducing manual inspections and providing automated data collection, mining companies can optimize their workflows and minimize downtime. This advanced system not only improves workplace safety standards but also contributes to a more productive and sustainable mining operation.

6.2 Limitations

Although the system achieves its intended objectives, there are certain limitations:

- The system performance depends on LoRa/network availability, and signal failure can delay emergency alerts.
- Sensor readings may be affected by harsh underground conditions like dust, heat, and moisture
- Continuous power supply and regular maintenance are required, which can be difficult in deep mining areas.
- The system may not function effectively in case of hardware damage or sudden power failure during mining operations.
- Initial installation and calibration of sensors may require skilled personnel and additional setup time.

6.3 Future Enhancements

To further improve the system, the following enhancements can be considered:

- Integration of GPS or UWB location tracking to precisely locate miners during emergencies.
- Addition of AI-based data analysis to predict hazards before they occur.
- Voice alert or vibration feedback directly to miners for immediate self-awareness.
- Battery health monitoring and automatic low-power alert system.
- Integration with mobile app or web dashboard for real-time remote monitoring by supervisors.

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Appendix A

Transmitter Side code

```
#include <Arduino.h>
#include "DHT.h"
#include <SPI.h>
#include <LoRa.h>
#include <MPU6050.h>
#include <LiquidCrystal_I2C.h>
#include "ThingSpeak.h"
#include <WiFi.h>
#define WIFI_SSID "MADAN"
#define WIFI_PASSWORD "12345678"
unsigned long myChannelNumber = 3139680;
const char *myWriteAPIKey = "PT5G1328D8ML199E";
WiFiClient client;
LiquidCrystal_I2C lcd(0x27, 16, 2);
MPU6050 mpu;
int16_t prevAccelX, prevAccelY, prevAccelZ;
unsigned long prevTime;
float fall_thresh = 2.0;
#define ss 5
#define rst 14
#define dio0 2
int counter = 0;
#define DHTPIN 17
#define DHTTYPE DHT11
DHT dht(DHTPIN, DHTTYPE);
float h, t;
int flame_pin = 16;
```

```
int flame_val = 0;
int smoke_pin = 34;
int smoke_val = 0;
int buzzer = 4;
int emergency_button = 33;
unsigned long button_time = 0;
unsigned long last_button_time = 0;
TaskHandle_t Task1;
TaskHandle_t Task2;
TaskHandle_t Task3;
unsigned long previousMillis = 0;
const long interval = 2000;
String fall_status = "No";
String smoke_status, flame_status;
bool emergency = false;
void IRAM_ATTR emergency_isr()
{
  button_time = millis();
  if (button_time - last_button_time > 1000)
  {
    emergency = true;
    last_button_time = button_time;
  }
}
void readDHT()
{
  h = dht.readHumidity();
  t = dht.readTemperature();
  Serial.print("Temperature = ");
  Serial.println(t);
  Serial.print("Humidity = ");
  Serial.println(h);
```

```
}
void showOnDisplay()
{
  lcd.clear();
  lcd.setCursor(0, 0);
  lcd.print("T:");
  lcd.print(t);
  lcd.print("C ");
  lcd.setCursor(9, 0);
  lcd.print("H:");
  lcd.print(h);
  lcd.print("%");
  lcd.setCursor(0, 1);
  lcd.print("Fi:");
  lcd.print(flame_status);
  lcd.setCursor(6, 1);
  lcd.print("S:");
  lcd.print(smoke_status);
  lcd.setCursor(11, 1);
  lcd.print("F:");
  lcd.print(fall_status);
}
// MARK: Read and send
void readSend(void *pvParameters)
{
  disableCore0WDT();
  Serial.print("Task1 running on core ");
  Serial.println(xPortGetCoreID());
  for (;;)
  {
    unsigned long currentMillis = millis();
    if (currentMillis - previousMillis >= interval)
```

```
{
  previousMillis = currentMillis;
  readDHT();
}
flame_val = digitalRead(flame_pin);
if (flame_val == 0)
{
  Serial.println("Flame Detected!");
  flame_status = "YES";
}
else
{
  flame_status = "NO";
}
smoke_val = analogRead(smoke_pin);
Serial.print("Smoke Sensor Value: ");
Serial.println(smoke_val);
if (smoke_val > 1400)
{
  smoke_status = "yes";
}
else
{
  smoke_status = "no";
}
if (t > 30 || smoke_val > 1400 || flame_val == LOW || emergency)
{
  digitalWrite(buzzer, HIGH);
}
else
{
  digitalWrite(buzzer, LOW);
```

```
    }
    LoRa.beginPacket(); // Send LoRa packet to receiver
    String data = data + (String)t + "," + (String)h + "," + (String)smoke_val + "," +
    Serial.println("Sending packet");
    LoRa.println(data);
    LoRa.endPacket();
    fall_status = "no";
    emergency = false;
    data = "";
    showOnDisplay();
    vTaskDelay(1000 / portTICK_PERIOD_MS);
  }
}
// MARK: Fall
void fallDetection(void *pvParameters)
{
  disableCore1WDT();
  Serial.print("Task2 running on core ");
  Serial.println(xPortGetCoreID());

  for (;;)
  {
    int16_t currAccelX, currAccelY, currAccelZ;
    mpu.getAcceleration(&currAccelX, &currAccelY, &currAccelZ);
    unsigned long currTime = millis();
    float deltaTime = (currTime - prevTime) / 1000.0; // Convert to seconds
    float accelChangeX = (currAccelX - prevAccelX) / 16384.0;
    float accelChangeY = (currAccelY - prevAccelY) / 16384.0;
    float accelChangeZ = (currAccelZ - prevAccelZ) / 16384.0;
    float accelRateX = accelChangeX / deltaTime;
    float accelRateY = accelChangeY / deltaTime;
    float accelRateZ = accelChangeZ / deltaTime;
```

```
    if (abs(accelRateX) > fall_thresh || abs(accelRateY) > fall_thresh || abs(accelRa
    {
      Serial.println("----------------Acceleration change exceeded 2g/s on at least o
      Serial.println("Fall Detected");
      fall_status = "yes";
    }
    Serial.print("Acceleration Change (g/s): ");
    Serial.print("X = ");
    Serial.print(accelRateX);
    Serial.print(", Y = ");
    Serial.print(accelRateY);
    Serial.print(", Z = ");
    Serial.println(accelRateZ);
    prevAccelX = currAccelX;
    prevAccelY = currAccelY;
    prevAccelZ = currAccelZ;
    prevTime = currTime;
    vTaskDelay(200 / portTICK_PERIOD_MS);
  }
}
// MARK: Upload
void upload(void *pvParameters)
{
  for (;;)
  {
    ThingSpeak.setField(1, t);
    ThingSpeak.setField(2, h);
    ThingSpeak.setStatus("Flame: "+flame_status + ", Smoke: " + smoke_status + ", Fal
    // ThingSpeak.setField(3, flame_status);
    // ThingSpeak.setField(4, smoke_status);
    // ThingSpeak.setField(5, fall_status);
    // Serial.println("------------uploading------------");
```

```
    // ThingSpeak.setField(1, 10);
    // write to the ThingSpeak channel
    int x = ThingSpeak.writeFields(myChannelNumber, myWriteAPIKey);
    if (x == 200)
    {
      Serial.println("[THING] Channel update successful.");
    }
    else
    {
      Serial.println("[THING] Problem updating channel. HTTP error code " + String(x)
    }
    vTaskDelay(pdMS_TO_TICKS(15000)); // Wait 1 second before next read
  }
}
// MARK: Setup
void setup()
{
  Serial.begin(115200);
  dht.begin();
  Serial.println("dht initialized");
  pinMode(flame_pin, INPUT);
  pinMode(smoke_pin, INPUT);
  pinMode(buzzer, OUTPUT);
  pinMode(emergency_button, INPUT_PULLUP);
  attachInterrupt(digitalPinToInterrupt(emergency_button), emergency_isr, FALLING);
  lcd.init();
  lcd.backlight();
  Serial.println();
  Serial.print("Connecting to ");
  Serial.println(WIFI_SSID);
  lcd.setCursor(0, 0);
  lcd.print("Connecting to ");
```

```
lcd.setCursor(0, 1);
lcd.print(WIFI_SSID);
WiFi.begin(WIFI_SSID, WIFI_PASSWORD);
while (WiFi.status() != WL_CONNECTED)
{
  delay(500);
  Serial.print(".");
}
Serial.println(".");
Serial.println("WiFi connected..!");
lcd.clear();
lcd.setCursor(0, 0);
lcd.print("WiFi connected..!");
delay(2000);
ThingSpeak.begin(client);
lcd.setCursor(0, 0);
lcd.print("initializing MPU...");
Wire.begin();
mpu.initialize();
mpu.CalibrateAccel();
mpu.setFullScaleAccelRange(MPU6050_ACCEL_FS_2);
Serial.println(mpu.testConnection() ? "MPU6050 connection successful" : "MPU6050 co
mpu.getAcceleration(&prevAccelX, &prevAccelY, &prevAccelZ);
lcd.clear();
lcd.setCursor(0, 0);
lcd.print("MPU6050 OK!");
delay(2000);
lcd.clear();
lcd.setCursor(0, 0);
lcd.print("initializing LoRa...");
LoRa.setPins(ss, rst, dio0);
while (!LoRa.begin(433E6))
```

```
{
  Serial.println(".");
  delay(500);
}
lcd.clear();
lcd.setCursor(0, 0);
lcd.print("LoRa Initialized");
delay(2000);
LoRa.setSyncWord(0xF3);
Serial.println("LoRa Initializing OK!");
xTaskCreatePinnedToCore(
    readSend, /* Task function. */
    "Task1",  /* name of task. */
    10000,    /* Stack size of task */
    NULL,     /* parameter of the task */
    1,        /* priority of the task */
    &Task1,   /* Task handle to keep track of created task */
    0);       /* pin task to core 0 */
delay(500);
xTaskCreatePinnedToCore(
    fallDetection, /* Task function. */
    "Task2",       /* name of task. */
    10000,         /* Stack size of task */
    NULL,          /* parameter of the task */
    1,             /* priority of the task */
    &Task2,        /* Task handle to keep track of created task */
    1);            /* pin task to core 0 */
delay(500);
xTaskCreatePinnedToCore(
    upload,  /* Task function. */
    "Task3", /* name of task. */
    10000,   /* Stack size of task */
```

```
      NULL,    /* parameter of the task */
      1,       /* priority of the task */
      &Task3,  /* Task handle to keep track of created task */
      0);      /* pin task to core 0 */
  delay(500);
}
void loop()
{
  // Serial.print("Sending packet: ");
  // Serial.println(counter);
  // LoRa.beginPacket();
  // LoRa.print("hello ");
  // LoRa.print(counter);
  // LoRa.endPacket();
  // counter++;
  // delay(5000);
}
```

Appendix B

Receiver side Code

```
#include <Arduino.h>
#include <SPI.h>
#include <LoRa.h>
#include <LiquidCrystal_I2C.h>
LiquidCrystal_I2C lcd(0x27, 16, 2);
#define ss D8
#define rst D3
String LoRaData;
float temp, humiduty;
int smoke, flame, emergency;
String fall, flame_status, smoke_status, emergency_status;
int buzzer = D0;
bool wait= false;
int button = D4;
void setup()
{
  Serial.begin(115200);
  Serial.println("LoRa Receiver");
  pinMode(buzzer, OUTPUT);
  pinMode(button, INPUT_PULLUP);
  lcd.init();
  lcd.backlight();
  lcd.setCursor(0, 0);
  lcd.print("initializing...");
  LoRa.setPins(ss, rst);
  while (!LoRa.begin(433E6))
  {
```

```
    Serial.println(".");
    delay(100);
  }
  lcd.clear();
  lcd.setCursor(0, 0);
  lcd.print("LoRa Initialized");
  LoRa.setSyncWord(0xF3);
  Serial.println("LoRa Initializing OK!");
  delay(2000);
}
String getValue(String data, char separator, int index)
{
  int found = 0;
  int strIndex[] = {0, -1};
  int maxIndex = data.length() - 1;
  for (int i = 0; i <= maxIndex && found <= index; i++)
  {
    if (data.charAt(i) == separator || i == maxIndex)
    {
      found++;
      strIndex[0] = strIndex[1] + 1;
      strIndex[1] = (i == maxIndex) ? i + 1 : i;
    }
  }
  return found > index ? data.substring(strIndex[0], strIndex[1]) : "";
}
void showOnDisplay()
{
  lcd.clear();
  lcd.setCursor(0, 0);
  lcd.print("T:");
  lcd.print(temp);
```

```
  lcd.print("C ");
  lcd.setCursor(9, 0);
  lcd.print("H:");
  lcd.print(humiduty);
  lcd.print("%");
  lcd.setCursor(0, 1);
  lcd.print("Fi:");
  lcd.print(flame_status);
  lcd.setCursor(6, 1);
  lcd.print("S:");
  lcd.print(smoke_status);
  lcd.setCursor(11, 1);
  lcd.print("F:");
  lcd.print(fall);
}
void loop()
{
  int packetSize = LoRa.parsePacket();
  if (packetSize)
  {
    Serial.print("Received packet '");
    while (LoRa.available())
    {
      LoRaData = LoRa.readString();
      Serial.print(LoRaData);
    }
    Serial.print("' with RSSI ");
    Serial.println(LoRa.packetRssi());
    String c = getValue(LoRaData, ',', 0);
    temp = c.toFloat();
    String h = getValue(LoRaData, ',', 1);
    humiduty = h.toFloat();
```

```
String s = getValue(LoRaData, ',', 2);
smoke = s.toInt();
if (smoke > 1200)
{
  smoke_status = "yes";
}
else
{
  smoke_status = "no";
}
String f = getValue(LoRaData, ',', 3);
flame = f.toInt();
if (flame == 0)
{
  flame_status = "YES";
}
else
{
  flame_status = "NO";
}
fall = getValue(LoRaData, ',', 4);
emergency = getValue(LoRaData, ',', 5).toInt();
Serial.print("Temperature: ");
Serial.println(temp);
Serial.print("Humidity: ");
Serial.println(humiduty);
Serial.print("Smoke: ");
Serial.println(smoke);
Serial.print("Flame: ");
Serial.println(flame);
Serial.print("Fall: ");
Serial.println(fall);
```

```
    if (temp > 30 || smoke > 1400 || flame == 0)
    {
      digitalWrite(buzzer, HIGH);
    }
    else
    {
      digitalWrite(buzzer, LOW);
    }
    if (emergency == 1)
    {
      wait = true;
    }
    else
    {
      showOnDisplay();
    }
  }
  if(wait)
  {
    lcd.clear();
    lcd.setCursor(0, 0);
    lcd.print("EMERGENCY!!!");
    lcd.setCursor(0, 1);
    lcd.print("HELP NEEDED!!!");
    digitalWrite(buzzer, HIGH);
    while(true)
    {
      if(digitalRead(button) == LOW)
      {
        wait = false;
        delay(1000);
        break;
```

```
      }
      delay(100);
    }
    digitalWrite(buzzer, LOW);
  }
}
```

Appendix C

Publications

My research paper titled **“[IoT Based Safety Helmet for Coal Minors]”** has been successfully published in the *International Journal of Scientific Research in Engineering and Management (IJSREM)*. The paper appears in **Volume** [**9**], **Issue** [**08**], [**2025**] after completing the journal’s peer-review and editorial evaluation process. I have also received the official publication certificate from IJSREM confirming the acceptance and publication of our work. This publication recognizes the academic relevance and practical contribution of our research.

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Madan U

In recognition to the publication of the paper titled

IOT Based Safety Helmet for Coal Minors

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IoT Based Safety Helmet For Coal Minors

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---***---

Abstract - Coal mine safety is a critical concern due to hazardous conditions like explosions, toxic gases, flooding, and collapses. Traditional monitoring and communication methods are inadequate, making modern solutions essential. The use of wireless networks, IoT devices, and sensor systems enables real-time monitoring of gas, temperature, humidity, and airflow. These technologies provide continuous data and instant alerts, ensuring faster response and improved miner safety.

Keywords: Smart helmet, coal mining, IoT, safety, environmental monitoring.

1. INTRODUCTION

Coal mining is a vital industry that meets global energy demands while supporting employment and GDP, particularly in developing economies. However, underground mining environments remain extremely hazardous, with risks such as gas explosions, toxic buildup, flooding, and rockfalls posing constant threats to workers' safety. Traditional safety measures like wired communication and manual reporting are increasingly seen as inadequate due to their maintenance challenges and delays in emergency response. To overcome these issues, modern mining operations are adopting wireless technologies such as Wi-Fi, LoRa, Zigbee, and 4G/5G, which provide more reliable and flexible communication. The integration of IoT-enabled safety helmets has further transformed mine safety by embedding sensors that monitor gas levels, temperature, humidity, and miners' real-time locations. This data is wirelessly transmitted to control centers for continuous monitoring and rapid hazard detection. In emergencies, the system triggers instant alerts, enabling faster rescue operations while also supporting predictive hazard prevention through analytics. By improving communication, situational awareness, and proactive safety measures, IoT-based smart helmets significantly reduce mining accidents and fatalities. Ultimately, such innovations modernize coal mining safety infrastructure, aligning with global trends toward smarter and more sustainable industrial practices.

2. LITERATURE REVIEW

The reviewed literature highlights significant advancements in the design and development of smart helmets aimed at improving miner safety. One work presents a sensor-integrated helmet with machine learning capabilities, using an Arduino microcontroller and Support Vector Machines (SVM) for hazard prediction. Another emphasizes a power-efficient smart helmet for monitoring both environmental conditions and miner health. Several studies propose helmets equipped with sensors to measure temperature, humidity, and hazardous gases such as methane and carbon monoxide, with ZigBee technology enabling real-time data transmission and emergency alert mechanisms like panic buttons. Other research also focuses on real-time monitoring of hazardous conditions, while some introduce early prediction and warning of potential calamities to improve safety measures. Further contributions integrate sensor networks and communication technologies for real-time monitoring and tracking of miners' health and location. Some works employ IoT and ARM Cortex-M for predictive alerts, reducing risks and ensuring rapid emergency response. Finally, one study demonstrates an IoT-enabled helmet capable of monitoring health and environmental conditions with up to 96% accuracy, significantly enhancing safety through early hazard detection and timely alerts. Collectively, these studies underline the growing role of IoT, wireless communication, and intelligent systems in creating proactive safety solutions for the mining industry.

3. METHODOLOGY

The proposed system is designed with two main units: a transmitter worn by miners and a receiver located in the control room. The transmitter integrates multiple sensors with an ESP32 microcontroller to continuously collect environmental and health-related data such as temperature, humidity, and toxic gas levels (e.g., benzene or ammonia). This ensures early detection of hazardous conditions like overheating, excessive air moisture, or toxic gas leaks. The collected data is processed by the ESP32 and transmitted using LoRa technology, which provides low-power, long-distance communication, making it highly effective in underground environments where traditional internet connectivity is unreliable. When sensor readings cross predefined safety thresholds, the system immediately triggers alerts, allowing quick response actions to prevent accidents and safeguard miners' lives. The receiver unit in the control room displays real-time data on an OLED screen, giving operators continuous visibility of underground conditions. Additionally, the data is sent to ThingSpeak, an IoT analytics platform, where it is aggregated, stored, and visualized through graphs and trends. This enables operators not only to monitor live data but also to analyze historical records, identify recurring risks, and plan preventive measures.

The software implementation uses Arduino IDE for programming and configuring the ESP32 and sensors, ensuring easy customization and flexibility. ThingSpeak serves as the cloud-based analytics platform, providing powerful tools for remote monitoring, pattern detection, and predictive safety analysis. Together, this combination of hardware and software creates a robust, efficient, and scalable solution that enhances miner safety, improves decision-making, and contributes to more reliable and sustainable mining operations.

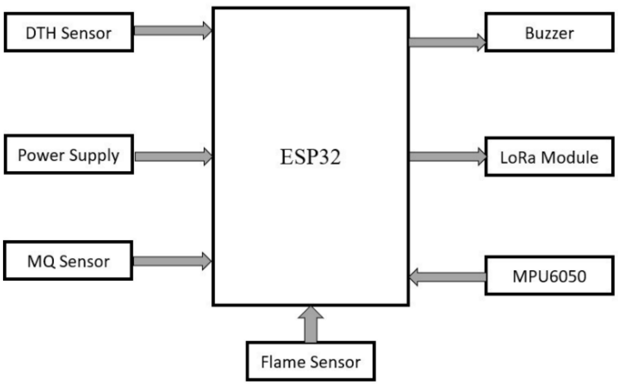


Fig -1: Block diagram of Transmitter End

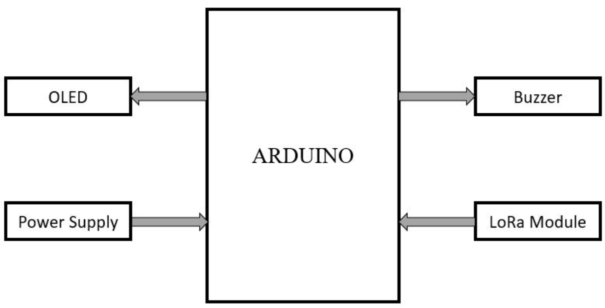


Fig -2: Block diagram of Receiver End

4. RESULTS – SYSTEM DEMONSTRATION

Fig -4(a): Hardware setup

All sensor data is processed by a microcontroller (ESP32/Arduino) and transmitted using LoRa, which is well-suited for underground mines due to its low power use and long range. The helmet continuously sends real-time data to the base station, enabling early detection of hazards and ensuring worker safety.

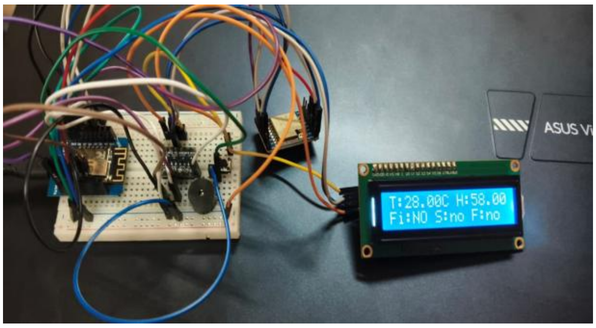


Fig -4(b): Prototype of receiver side

The base station monitors mine conditions in real time using an Arduino Uno with a LoRa module that receives data from the helmet transmitter. It processes temperature, humidity, air quality, and flame detection signals, triggering LED alerts when hazards are detected. The system can be expanded with an LCD display or cloud-based IoT platform for remote monitoring, ensuring quick response and improved worker safety.

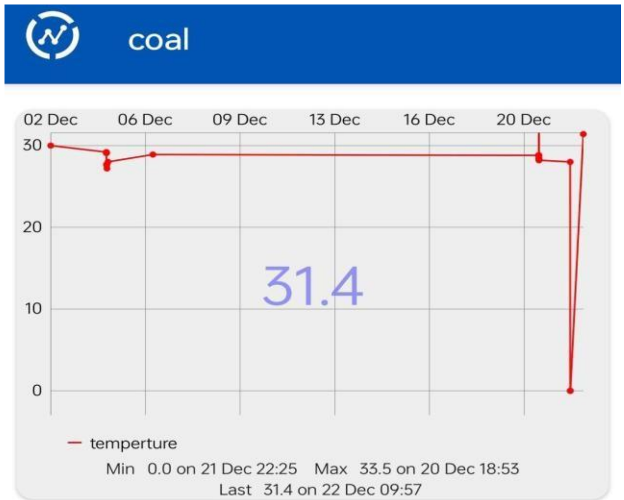
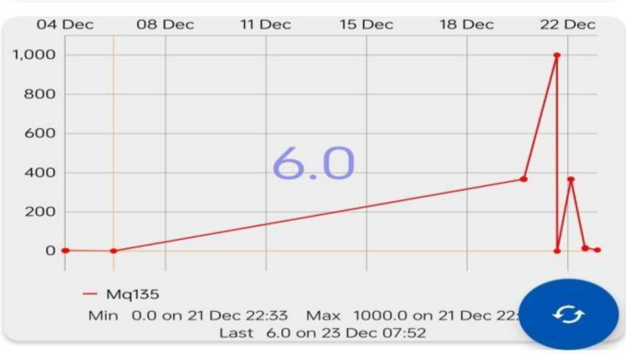


Fig -4(c): Temperature data displayed in cloud-based IoT platform

Fig -4(d): Humidity data displayed in cloud-based



IoT platform

Fig -4(e): MQ135 gas sensor data displayed in cloud-based IoT platform

The IoT-based coal mine safety system monitors temperature, humidity, and gas levels in real time. Readings showed stable values but sudden drops to 0 on December 21 indicated sensor or transmission faults. Humidity remained high at 95%, posing corrosion and air quality risks, while gas concentration spiked to 1000 PPM, suggesting a possible leak. The system is effective but needs better sensor reliability, data stability, and automated alerts for improved safety.

5. CONCLUSIONS

Wireless networks in coal mines enable real-time monitoring of gas, temperature, and humidity for early hazard detection. They ensure reliable communication between miners and control centers, support quick emergency response, and provide automated alerts and tracking for rescue operations. In addition to improving safety, these systems reduce manual inspections, optimize workflows, and enhance overall efficiency

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